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Design of Digital Filters for Frequency Weightings Required for Risk Assessments of Workers Exposed to Vibration

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Abstract: Many workers are exposed to vibration in their industrial environment. Vibration can be transmitted through a vehicle seat or a hand-held power tool. Excessive vibration exposure may cause health problems and therefore it is important that the worker's vibration exposure is assessed, which may require measurement by the equipment manufacturer or the employer. Human exposure to vibration may be measured using accelerometers; however, weighting filters are required to correlate the physical vibration measurements to the human's response to vibration. ISO 2631, BS 6841 and ISO 5349-1 describe suitable weighting filters, but do not explain how to implement them for digitally recorded acceleration data. ISO 8041 Annex C suggests a method but does not provide a solution. By using the bilinear transform, it is possible to transform the analogue equations given in the standards into digital filters. This paper describes the implementation of the weighting filters as digital IIR (Infinite Impulse Response) filters and provides all the necessary formulae to directly calculate the filter coefficients for any sampling frequency. Thus, the filters in the standards can be implemented in any numerical software.

Key words: ISO 8041, ISO 2631, ISO 5349, BS 6841, Occupational health and safety, Whole-body and hand-arm vibration, Frequency weighting, Digital filter

Introduction

Humans are exposed to vibration from many different sources, including product or passenger transportation systems, military vehicles, agricultural machines, and hand-held power tools. Prolonged exposure may lead to motion sickness, discomfort or health-related problems such as low back pain or hand-arm vibration syndrome[1–4]. Limits on the amount of daily vibration exposure to which workers may be exposed were introduced across Europe in 2005[5] and are being enforced by member states (for example, in the UK, the directive is being enforced through the Control of Vibration at Work Regulations[6]), similar legislation has followed in the USA for hand-arm vibration and it is expected that similar legislation will be introduced in Japan in the future. Exposure limits can be expressed as the maximum time that an employee may be exposed to whole-body or hand-arm vibration before reaching an Exposure Action Value (where the employer should take further action to reduce the vibration exposure to the employee) or Exposure Limit Value (where the employee should stop using vibration emitting equipment). The exposure might come from a single machine or tool, or from a combination of machines or tools used at different times during the day. The limits, combined with health surveillance, worker training and generic minimisation of risks, are designed to eliminate vibration-induced injury within the workplace.

It is the responsibility of equipment manufacturers to produce documentation to show the vibration emission from their machines or tools. Standard test codes exist for most tool types, where the frequency-weighted vibration must be measured for the tool in a particular configuration and operated

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Table 1. Frequency weightings used in ISO 2631-1, ISO 2631-2, ISO 5349-1 and BS 6841, and contexts of use

Filter	Direction	Primary context of use	Standard
W_b	Vertical	Seat vibration	BS 6841
W_c	Fore-aft	Backrest vibration	ISO 2631-1, BS 6841
W_d	Fore-aft and lateral	Seat vibration	ISO 2631-1, BS 6841
W_e	Roll, pitch, yaw	Rotational seat vibration	ISO 2631-1, BS 6841
W_f	Vertical	Motion sickness	ISO 2631-1, BS 6841
W_g	Vertical	Activity interference	BS 6841
W_h	Fore-aft, lateral and vertical	Hand-arm vibration	ISO 5349-1
W_j	Vertical	Head vibration	ISO 2631-1
W_k	Vertical	Seat vibration	ISO 2631-1
W_m	All	Building vibration	ISO 2631-2

in a particular way. The configuration in these test codes rarely replicate real-use conditions. A large number of occupational health professionals, employers, manufacturers and consultants are therefore required to make measurements of whole-body and/or hand-transmitted vibration exposure, in order to understand the nature of the vibration that the worker is exposed to, and to estimate health risks.

Vibration measurement for assessment of human exposure is usually made by placing accelerometers on the vibrating surface (e.g. on the seat surface between the operator and the seat for whole-body vibration measurements or on the handle of a power tool for hand-transmitted vibration measurements). A typical accelerometer is calibrated to produce 1 Volt for each g (9.81 m/s^2) of acceleration it is exposed to. The effect of vibration on a human is a function of frequency and therefore it is essential that accelerometers are capable of providing reliable data at the most important frequencies. Band-limiting filters in ISO 8041[7] occur at 0.4 Hz and 100 Hz (whole-body vibration filters), at 0.79 Hz and 100 Hz (building vibration filter), at 6.3 Hz and 1,259 Hz (hand-arm vibration filter) and at 0.08 Hz and 0.63 Hz (motion sickness filter), and therefore measurement equipment should be selected to have a nominally linear response at least within the relevant frequency ranges. Human response to vibration is a function of frequency, and therefore measured data should be "weighted" to give greater prominence to frequencies where humans are most sensitive[8,9]. It is necessary to use standard methods for frequency weighting so that results from different measurements, using different equipment can be compared. Three standards are in common use: ISO 2631[10] and BS 6841[11] are used for whole-body vibration measurements and ISO 5349-1[12] for hand-transmitted vibration measurements. ISO 2631 and BS 6841 describe weighting filters for combinations of standing, sitting and recumbent subjects for the analysis of health, discomfort, perception and motion sickness. In addition, they describe how different weighting filters should be used for different axes of measurement (fore-aft, lateral, vertical, rotational). There are six weightings used in each of ISO 2631-1 and BS 6841, of which four are common to both. One weighting is given in ISO 2631-2 and also one weighting in ISO 5349-1. In total, there are ten weightings defined in the standards (Table 1, Fig. 1). The standards present the frequency-domain weighting as one-third octave band multiplication factors and as analogue equations. The one-third octave factors may be used if the recorded data are in the frequency-domain and divided into one-third octave bands. However, this is not usual as most recorded vibration (i.e. acceleration) data is obtained and stored digitally in the time-domain as a series of sampled values, either on a computer or a digital data logger. ISO 2631, BS 6841 and ISO 5349-1 do not explain how to apply the given weighting filters to digital time-domain data. ISO 8041 suggests the use of either the bilinear transform or impulse invariant method, but does not provide a solution, thus making it difficult for professionals without the necessary signal-processing expertise to implement the weighting filters. ISO 8041 also includes some MATLAB® code for calculating the coefficients of W_k; however, this necessitates the purchase of a commercial product for implementation, and, if the analysis software were to be written in a programming language other than MATLAB®, each filter would need to be explicitly designed for each sampling frequency that the final user might require.

Zuo and Nayfeh[13] produced approximations of three of the ISO 2631-1 weighting curves for implementation in the digital domain, but did not present the digital filter coefficients. Osama and Guan[14] had previously presented

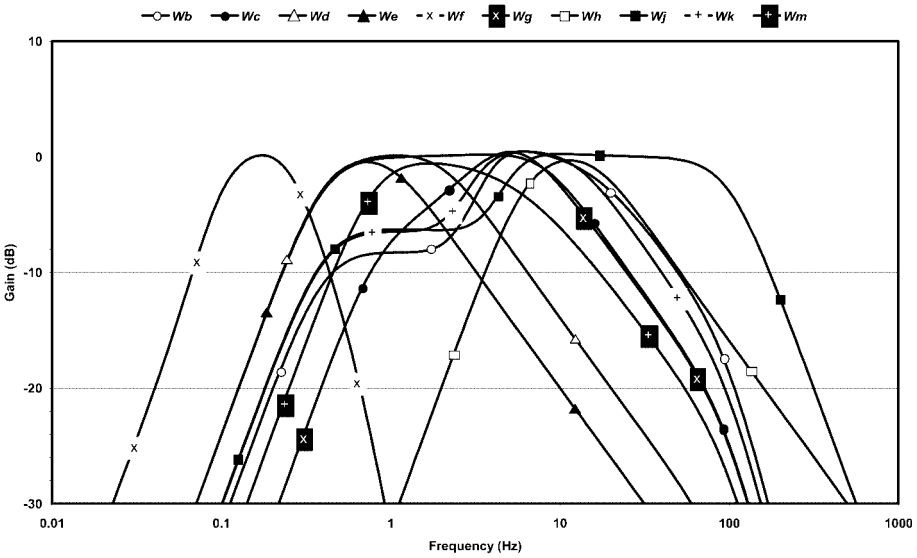


Fig. 1. Moduli of the frequency weightings used in ISO 2631-1, ISO 2631-2, BS 6841 and ISO 5349-1.

a FORTRAN program to calculate the coefficients of the BS 6841 W_b and W_d weighting filters using a similar technique to that described here. This paper describes the implementation of all of the weighting filters defined in ISO 2631-1, ISO 2631-2, BS 6841 and ISO 5349-1 as digital IIR filters, and provides the necessary simple formulae to directly calculate the filter coefficients for any sampling frequency, thus removing the requirement to know the sampling frequency before run-time (i.e. the filters do not need to be calculated in advance using third-party software for each possible sampling frequency).

Frequency Weighting Filters

The filters defined in ISO 2631 and BS 6841 comprise a number of sections, which are then cascaded in different combinations to produce the set of weighting filters. Different filters are used depending on the measurement scenario and application as defined in the standard (for example, ISO 2631-1 weighting filter W_k is to be used when assessing z-axis vibration comfort for a seated person).

ISO 2631-1 and ISO 2631-2

The component filter sections are defined in ISO 2631 and ISO 8041 as analogue transfer functions. Band-limiting is performed using filters labelled as 'high-pass' and 'low-pass'; other component parts of the process are achieved using 'acceleration-velocity transition' and 'upwards step' filters. These can be expressed in the s-domain as follows:

High-pass filter

$$H_h(s) = \frac{s^2}{s^2 + \frac{\omega_1}{Q_1}s + {\omega_1}^2} \quad (1)$$

Low-pass filter

$$H_l(s) = \frac{{\omega_2}^2}{s^2 + \frac{\omega_2}{Q_2}s + {\omega_2}^2} \quad (2)$$

Acceleration-velocity transition filter

$$H_t(s) = \frac{\frac{{\omega_4}^2}{\omega_3}s + {\omega_4}^2}{s^2 + \frac{\omega_4}{Q_4}s + {\omega_4}^2} \quad (3)$$

Upward step filter

$$H_s(s) = \frac{s^2 + \frac{{\omega_5}^2}{Q_5}s + {\omega_5}^2}{s^2 + \frac{\omega_6}{Q_6}s + {\omega_6}^2} \quad (4)$$

The total weighting filter is defined as the combination of Equations 1 to 4 as shown in Table 2. The numeric values to be used are shown in Table 3.

BS 6841

For BS 6841, the component filter sections are defined in the analogue, Laplace, s-domain. Band-limiting, high-pass and low-pass filters are combined into a single 'bandpass' filter; two other expressions are provided in the standard for the weighting filter, the relevant one of which is selected according to the frequency-weighting being implemented. The

Table 2. Weighting filters for ISO 2631 created from cascading filter sections

Filter	Filter stages	Filter	Filter stages
W_c	$H_h(s) \cdot H_l(s) \cdot H_t(s)$	W_f	$H_h(s) \cdot H_l(s) \cdot H_t(s) \cdot H_s(s)$
W_d	$H_h(s) \cdot H_l(s) \cdot H_t(s)$	W_j	$H_h(s) \cdot H_l(s) \cdot H_s(s)$
W_e	$H_h(s) \cdot H_l(s) \cdot H_t(s)$	W_k	$H_h(s) \cdot H_l(s) \cdot H_t(s) \cdot H_s(s)$
W_m	$H_h(s) \cdot H_l(s) \cdot H_t(s)$		

Table 3. Numeric values to be used in Equations 1 to 4. Note that $\omega = 2 \cdot \pi \cdot f$

	H_h		H_l		H_t			H_s			
	f_1 (Hz)	Q_1	f_2 (Hz)	Q_2	f_3 (Hz)	f_4 (Hz)	Q_4	f_5 (Hz)	Q_5	f_6 (Hz)	Q_6
W_c	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	8	8	0.63	–	–	–	–
W_d	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	2	2	0.63	–	–	–	–
W_e	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	1	1	0.63	–	–	–	–
W_f	0.08	$1/\sqrt{2}$	0.63	$1/\sqrt{2}$	$f_s/2$	0.25	0.86	0.06	0.8	0.1	0.8
W_j	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	–	–	–	3.75	0.91	5.3	0.91
W_k	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	12.5	12.5	0.63	2.37	0.91	3.3	0.91
W_m	0.7943	$1/\sqrt{2}$	100	$1/\sqrt{2}$	5.68	5.684	0.5	–	–	–	–

Table 4. Weighting filters for BS 6841 created from cascading filter sections

Filter	Filter stages	Filter	Filter stages
W_{b_BS}	$H_{b_BS}(s) \cdot H_{w2_BS}(s)$	W_{e_BS}	$H_{b_BS}(s) \cdot H_{w1_BS}(s)$
W_{c_BS}	$H_{b_BS}(s) \cdot H_{w1_BS}(s)$	W_{f_BS}	$H_{b_BS}(s) \cdot H_{w2_BS}(s)$
W_{d_BS}	$H_{b_BS}(s) \cdot H_{w1_BS}(s)$	W_{g_BS}	$H_{b_BS}(s) \cdot H_{w1_BS}(s)$

filters can be expressed as follows (expressed here as a function of ω, with indices changed to remain unique within this paper):

Band-pass filter:

$$H_{b_BS}(s) = \frac{s^2}{s^2 + \frac{\omega_{1_BS}}{Q_{1a_BS}} s + {\omega_{1_BS}}^2} \cdot \frac{{\omega_{2_BS}}^2}{s^2 + \frac{\omega_{2_BS}}{Q_{1b_BS}} s + {\omega_{2_BS}}^2} \quad (5)$$

Weighting filter 1, for W_c, W_d, W_e and W_g:

$$H_{w1_BS}(s) = \frac{s + \omega_{3_BS}}{s^2 + \frac{\omega_{4_BS}}{Q_{2_BS}} s + {\omega_{4_BS}}^2} \cdot \frac{k\,{\omega_{4_BS}}^2}{\omega_{3_BS}} \quad (6)$$

Weighting filter 2, for W_b and W_f:

$$H_{w2_BS}(s) = \frac{s + \omega_{3_BS}}{s^2 + \frac{\omega_{4_BS}}{Q_{2_BS}} s + {\omega_{4_BS}}^2} \cdot \frac{s^2 + \frac{\omega_{5_BS}}{Q_{3_BS}} s + {\omega_{5_BS}}^2}{s^2 + \frac{\omega_{6_BS}}{Q_{4_BS}} s + {\omega_{6_BS}}^2} \cdot \frac{k{\omega_{4_BS}}^2\,{\omega_{6_BS}}^2}{\omega_{3_BS}\,{\omega_{5_BS}}^2} \quad (7)$$

The complete filter is produced by cascading the band-pass filter and appropriate weighting filter (Table 4).

It can be shown that an equivalence exists between the equations for the ISO 2631-1 weighting filters (Equations 1 to 4) and the equations for the BS 6841 weighting filters (Equations 5 to 7), such that Equations 1 to 4 may be used to calculate the BS 6841 weighting filters. Table 5 gives the parameters required to implement the BS 6841 weighting filters using Equations 1 to 4. The individual filter stages are then cascaded as shown in Table 6.

ISO 5349-1

The weighting filter W_h used in ISO 5349-1 is also defined in ISO 8041. Filtering is performed using the same two-stage process as for the previously considered functions. Band-limiting is performed using high and low-pass filters as defined in equations (1) and (2).

The frequency-weighting part of the filter is defined in the s-domain as follows (expressed here as a function of ω, with the indices changed to remain unique within this paper):

$$H_w(s) = \frac{(s + \omega_7) \cdot {\omega_8}^2}{\left[s^2 + \frac{\omega_8}{Q_7} s + {\omega_8}^2\right] \omega_7} \quad (8)$$

Numeric values for the band-limiting and weighting filter coefficients are provided in Table 7. The high-pass and low-pass filters are cascaded with the frequency-weighting filter to produce the overall weighting filter W_h, where:

$$W_h = H_h(s) \cdot H_l(s) \cdot H_w(s) \quad (9)$$

Table 5. Numeric values required to implement BS 6841 using the equations for ISO 2631-1 (Equations 1 to 4)

	H_h		H_l		H_t			H_s				
	f_1 (Hz)	Q_1	f_2 (Hz)	Q_2	f_3 (Hz)	f_4 (Hz)	Q_4	f_5 (Hz)	Q_5	f_6 (Hz)	Q_6	G
W_{b_BS}	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	16	16	0.55	2.5	0.9	4	0.95	1.15
W_{c_BS}	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	8	8	0.63	–	–	–	–	1.00
W_{d_BS}	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	2	2	0.63	–	–	–	–	1.00
W_{e_BS}	0.4	$1/\sqrt{2}$	100	$1/\sqrt{2}$	1	1	0.63	–	–	–	–	1.00
W_{f_BS}	0.08	$1/\sqrt{2}$	0.63	$1/\sqrt{2}$	$f_s/2$	0.25	0.86	0.0625	0.8	0.1	0.8	1.024
W_{g_BS}	0.8	$1/\sqrt{2}$	100	$1/\sqrt{2}$	1.5	5.3	0.68	–	–	–	–	0.42

$\omega = 2 \cdot \pi \cdot f$

Table 6. Weighting filters for BS 6841 created from cascading filter sections calculated using Equations 1 to 4

Filter	Filter stages	Filter	Filter stages
W_{b_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s) \cdot H_s(s)$	W_{e_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s)$
W_{c_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s)$	W_{f_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s) \cdot H_s(s)$
W_{d_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s)$	W_{g_BS}	$G \cdot H_h(s) \cdot H_l(s) \cdot H_t(s)$

The values of gain, G, are taken from Table 5.

IIR Filter Design Using the Bilinear Transform

All modern data acquisition systems used for recording acceleration for analysis of human exposure are digital systems. They generally store recorded data (typically from three accelerometers (one for each axis)) into non-volatile memory that can be transferred to a PC for analysis at a later date. The first stage of analysis using ISO 2631, BS 6841 or ISO 5349 is to apply the weighting filter to the data. As the data is in the time-domain, and as subsequent processing is most straight forward if carried out in the time-domain, it is ideal to perform such filtering in the time-domain too. Some metrics defined in the standards (e.g. crest factor or VDV[a]) require a time-domain solution. An important parameter in the digital data acquisition process, is the selection of an appropriate sampling rate. The Nyquist-Shannon sampling theorem states that the sampling frequency should not be less than twice the maximum frequency that is to be recorded. In practice, with the filters described in this paper, the minimum sampling frequency needs to be higher than this to ensure that the frequency magnitude and phase responses are within the tolerances given in ISO 8041 and BS 6841. The minimum sampling frequency required to meet the required tolerances are given in Table 9.

[a]VDV (Vibration Dose Value) is defined as $\sqrt[4]{\int_0^T a_w^4(t)\,dt}$, and is a metric which emphasises shocks more than an r.m.s. measurement would[8].

Table 7. Band-limiting and filter values to be used for the filter used in ISO 5349, cascading equations 1, 2, and 8

f_1 (Hz)	Q_1	f_2 (Hz)	Q_2	f_7 (Hz)	f_8 (Hz)	Q_8
6.31	$1/\sqrt{2}$	1258.9	$1/\sqrt{2}$	15.915	15.915	0.64

$\omega = 2 \cdot \pi \cdot f$

IIR digital filters

With a digital filter it is possible to modify the frequency and phase response of a system by applying a series of multiplications and additions to the time-domain data (for a detailed description of digital filtering see Oppenheim and Schafer[15] or Rabiner and Gold[16]). There are essentially two methods of filtering a signal digitally: Finite Impulse Response (FIR) and Infinite Impulse response (IIR). FIR filters work by multiplying the current and previous data samples by a set of coefficients and summing the results. The value of the current output sample, $y[n]$, is given by:

$$y[n] = \sum_{k=0}^{M} b_k \cdot x[n-k] \tag{10}$$

where b_k is the kth filter coefficient and $x[n-1]$ is the

Table 8. IIR filter coefficients for weighting filter sections H_h, H_l, H_t, H_s and H_w

	H_h	H_l	H_t
a_0	$4Q_1 + 2\omega_1' + \omega_1'^2$	$4Q_2 + 2\omega_2' + \omega_2'^2 Q_2$	$4Q_4 + 2\omega_4' + \omega_4'^2 Q_4$
a_1	$2\omega_1'^2 - 8Q_1$	$2\omega_2'^2 Q_2 - 8Q_2$	$2\omega_4'^2 Q_4 - 8Q_4$
a_2	$4Q_1 - 2\omega_1' + \omega_1'^2$	$4Q_2 - 2\omega_2' + \omega_2'^2 Q_2$	$4Q_4 - 2\omega_4' + \omega_4'^2 Q_4$
b_0	$4Q_1$	$\omega_2'^2 Q_2$	$\omega_4'^2 Q_4 + 2\frac{Q_4\,\omega_4'^2}{\omega_3'}$
b_1	$-8Q_1$	$2\omega_2'^2 Q_2$	$2\omega_4'^2 Q_4$
b_2	$4Q_1$	$\omega_2'^2 Q_2$	$\omega_4'^2 Q_4 - 2\frac{Q_4\,\omega_4'^2}{\omega_3'}$

	H_s	H_w
a_0	$\frac{4Q_6 + 2\omega_6' + \omega_6'^2 Q_6}{Q_5}$	$\omega_7'\,Q_7\,\omega_8'^2 + 2\omega_7'\,\omega_8' + 4\omega_7'\,Q_7$
a_1	$\frac{2\omega_6'^2 Q_6 - 8Q_6}{Q_5}$	$2\omega_7'\,Q_7\,\omega_8'^2 - 8\omega_7'\,Q_7$
a_2	$\frac{4Q_6 - 2\omega_6' + \omega_6'^2 Q_6}{Q_5}$	$\omega_7' Q_7\,\omega_8'^2 - 2\omega_7'\,\omega_8' + 4\omega_7'\,Q_7$
b_0	$\frac{4Q_5 + 2\omega_5' + \omega_5'^2 Q_5}{Q_6}$	$\omega_7'\,Q_7\,\omega_8'^2 + 2Q_8\,\omega_8'^2$
b_1	$\frac{2\omega_5'^2 Q_5 - 8Q_5}{Q_6}$	$2\omega_7'\,Q_7\,\omega_8'^2$
b_2	$\frac{4Q_5 - 2\omega_5' + \omega_5'^2 Q_5}{Q_6}$	$\omega_7'\,Q_7\,\omega_8'^2 - 2Q_8\,\omega_8'^2$

Numeric values for ω and Q may be taken from Tables 3, 5 & 7 where ω is warped to ω' using Equation 13. The filter sections are combined for the desired weighting filter using Table 2 (ISO 2631-1) or Table 6 (BS 6841). ISO 5349 cascades band-limiting and weighting elements using Equation 9.

Table 9. The minimum sample rate required for each filter to meet the tolerances specified in ISO 8041 and BS 6841

Filter	f_s *min (Hz)*	*Filter*	f_s *min (Hz)*
BS W_b	$12\,f_2$	ISO W_c	$9\,f_2$
BS W_c	$12\,f_2$	ISO W_d	$9\,f_2$
BS W_d	$12\,f_2$	ISO W_e	$9\,f_2$
BS W_e	$12\,f_2$	ISO W_f	$11\,f_2$
BS W_f	$12\,f_2$	ISO W_h	$9\,f_2$
BS W_g	$12\,f_2$	ISO W_j	$8\,f_2$
		ISO W_k	$9\,f_2$
		ISO W_m	$9\,f_2$

Sampling frequency is presented in the format used in ISO 8041 C.2.4, where it is presented as a multiple of the upper frequency limit, f_2, as specified in Tables 3, 5 and 7.

previous input sample. An IIR filter uses previous output values in addition to previous input values to calculate the current output sample value, which may be given by:

$$y[n] = \frac{1}{a_0}\left(\sum_{j=0}^{M} b_j \cdot x[n-j] - \sum_{k=1}^{M} a_k \cdot y[n-k]\right) \qquad (11)$$

An IIR filter is recursive and requires less filter coefficients than the equivalent FIR filter; however, IIR filter design is more difficult and a poor design may result in an unstable filter, whereas an FIR filter is always stable.

Bilinear transform

There are a number of methods for deriving a digital filter from an analogue s-domain filter. In this paper, the authors use the bilinear transform method with frequency warping, as it maps the entire $j\omega$-axis in the s-plane to one revolution of the unit circle in the z-plane[15]. In the bilinear transform method of digital IIR filter design, s in the analogue s-domain equation is replaced by the bilinear transform defined by Equation 12[16].

$$s \rightarrow 2\,\frac{(1-z^{-1})}{(1+z^{-1})} \qquad (12)$$

There is, however, a non-linear relationship between the analogue frequency and the digital frequency. Pre-warping the frequencies used in the analogue s-domain equations (with the substitution shown in Equation 13) can eliminate this problem.

$$\omega_n' \rightarrow 2 \tan\left[\frac{\omega_n}{2}\right] \quad (13)$$

where ω_n is the normalised filter design frequency (i.e. $2\pi\omega_c/\omega_s$ or $2\pi f_c/f_s$, where ω_c and f_c are the centre frequency and ω_s and f_s are the sampling frequency) and ω_n' is the normalised warped frequency. To design the equivalent digital IIR filter, Equation 12 is substituted into the analogue s-domain equation, which is simplified until it is in the form of Equation 14. The filter coefficients (a_0 to a_2 and b_0 to b_2) may then be read directly. Lastly, the values for ω_n should be warped using Equation 13 to give the final filter coefficients.

$$H(z) = \frac{b_2 z^{-2} + b_1 z^{-1} + b_0}{a_2 z^{-2} + a_1 z^{-1} + a_0} \quad (14)$$

Worked example: Low-pass filter

This section contains a worked example to show the process of designing an IIR digital filter from an analogue s-domain equation. Consider Equation 2 which may be generalised as:

$$H(s) = \frac{\omega^2}{s^2 + \frac{\omega}{Q}s + \omega^2} \quad (15)$$

and let $\omega = 1570.796$ rad/s (i.e. $f = 250$ Hz), $Q = 1/\sqrt{2}$, and $f_s = 1{,}000$ Hz.

Stage one of the process is to apply the bilinear transform. Substituting Equation 12 into Equation 15 gives:

$$H(z) = \frac{\omega^2}{\left[2\frac{(1-z^{-1})}{(1+z^{-1})}\right]^2 + \frac{\omega}{Q}\left[2\frac{(1-z^{-1})}{(1+z^{-1})}\right] + \omega^2} \quad (16)$$

The second stage of the process is to expand and simplify Equation 16, to place it in the form of Equation 14:

$$H(z) = \frac{[\omega^2 Q]z^{-2} + [2\omega^2 Q]z^{-1} + [\omega^2 Q]}{[4Q + \omega^2 Q - 2\omega]z^{-2} + [2\omega^2 Q - 8Q]z^{-1} + [4Q + \omega^2 Q + 2\omega]} \quad (17)$$

The third stage is to perform the frequency warping procedure. The normalised centre frequency, $\omega_n = 2\pi\omega / \omega_s = 2\pi f/f_s = 2\pi \times 250/1000 = 1.57$ rad/s. From Equation 13, the warped frequency $\omega_n' = 2 \tan\left[\frac{\omega_n}{2}\right] = 2.0$ rad/s.

Thus, the IIR filter coefficients are as follows (where $\omega_n' = 2.0$ and $Q = 1/\sqrt{2}$):

$b_2 = \omega_n'^2 Q = 2.83$

$b_1 = \omega_n'^2 Q = 5.66$

$b_0 = \omega_n'^2 Q = 2.83$

$a_2 = 4Q + \omega_n'^2 Q - 2\omega_n' = 1.66$

$a_1 = 2\omega_n'^2 Q - 8Q = 0$

$a_0 = 4Q + \omega_n'^2 Q + 2\omega_n' = 9.66$

Once a user has calculated these six quantities, they can be used with Equation 11 in order to calculate the filtered acceleration. In order to complete the full frequency weighting, each stage will need to be completed, using the output from each stage as the input to the next.

IIR Digital Weighting Filter Coefficients

By applying the bilinear transform to Equations 1 to 4 and Equation 8 it is possible to design the IIR weighting filter with the corresponding frequency response. The general form for an IIR filter section is given by Equation 14, where a_0 to b_2 are the coefficients defined in Table 8, which uses the numeric values given in Tables 3, 5 and 6. The values of ω from these tables need to be warped using Equation 13. The filter sections are combined to give the desired weighting filter as shown in Table 2 (ISO 2631) and Table 6 (BS 6841). For ISO 5349-1 W_h is calculated using Equation 9.

As stated in Section 3, it is necessary to specify a minimum sample rate to be used in order for the filters to meet the tolerances specified in the standards. Note that this is the sample rate of the data as it is filtered, it is possible to up-sample recorded data to a suitable sample rate for filtering through interpolation, however, it is advisable to record the data at a sample rate which is not less than the minimum stated in this section. Table 9 gives the minimum sample rate for each of the weighting filters, required to meet the tolerances specified in the standards.

Conclusions

Many workers are exposed to vibration in their working environment, through a vehicle seat or a hand-held power tool. Excessive vibration exposure may cause health problems and therefore it is often important that the worker's vibration exposure is measured and/or monitored by the equipment manufacturer or by the employer. Human exposure to vibration may be measured using accelerometers; however, weighting filters are required to correlate the physical vibration measurements to the human's response to vibration. The weighting filters are presented in the relevant standards (ISO 2631, BS6841 and ISO 5349-1), but not in a format that can be implemented directly. This paper has presented a method of implementing the frequency weightings as digital filters, which can be used with any numerical software. Using the information presented, it is possible to design an IIR digital filter of arbitrary sample

rate to weight the recorded time-domain vibration data, necessary for calculating the weighted r.m.s. level (and thus the maximum health-related exposure times) in accordance with the international standards and the EU Directive.

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